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#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main()
{
    char str[1000], token[100];
    int i = 0, j;

    printf("Enter C program (end with $):\n");

    /* Read input until $ */
    while ((str[i] = getchar()) != '$' && i < 999)
        i++;
    str[i] = '\0';

    printf("\nTokens:\n");

    i = 0;

    while (str[i] != '\0')
    {
        /* Ignore spaces, tabs and new lines */
        if (isspace(str[i]))
        {
            i++;
            continue;
        }

        /* Ignore single-line comments */
        if (str[i] == '/' && str[i + 1] == '/')
        {
            i += 2;
            while (str[i] != '\0' && str[i] != '\n')
                i++;
            continue;
        }

        /* Ignore multi-line comments */
        if (str[i] == '/' && str[i + 1] == '*')
        {
            i += 2;
            while (str[i] != '\0' &&
                !(str[i] == '*' && str[i + 1] == '/'))
                i++;

            if (str[i] != '\0')
                i += 2;

            continue;
        }

        /* Identify identifiers */
        if (isalpha(str[i]) || str[i] == '_')
        {
            j = 0;

            while (isalnum(str[i]) || str[i] == '_')

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        {
            if (j < 99)
                token[j++] = str[i];
            i++;
        }

        token[j] = '\0';

        printf("IDENTIFIER : %s\n", token);
        continue;
    }

    /* Identify constants */
    if (isdigit(str[i]))
    {
        j = 0;

        while (isdigit(str[i]) || str[i] == '.')
        {
            if (j < 99)
                token[j++] = str[i];
            i++;
        }

        token[j] = '\0';

        printf("CONSTANT : %s\n", token);
        continue;
    }

    /* Identify operators */
    if (strchr("+-*/%=<>!&|", str[i]))
    {
        if ((str[i] == '+' && str[i + 1] == '+') ||
            (str[i] == '-' && str[i + 1] == '-') ||
            (str[i] == '=' && str[i + 1] == '=') ||
            (str[i] == '!' && str[i + 1] == '=') ||
            (str[i] == '<' && str[i + 1] == '=') ||
            (str[i] == '>' && str[i + 1] == '=') ||
            (str[i] == '&' && str[i + 1] == '&') ||
            (str[i] == '|' && str[i + 1] == '|'))
        {
            printf("OPERATOR : %c%c\n", str[i], str[i + 1]);
            i += 2;
        }
        else
        {
            printf("OPERATOR : %c\n", str[i]);
            i++;
        }

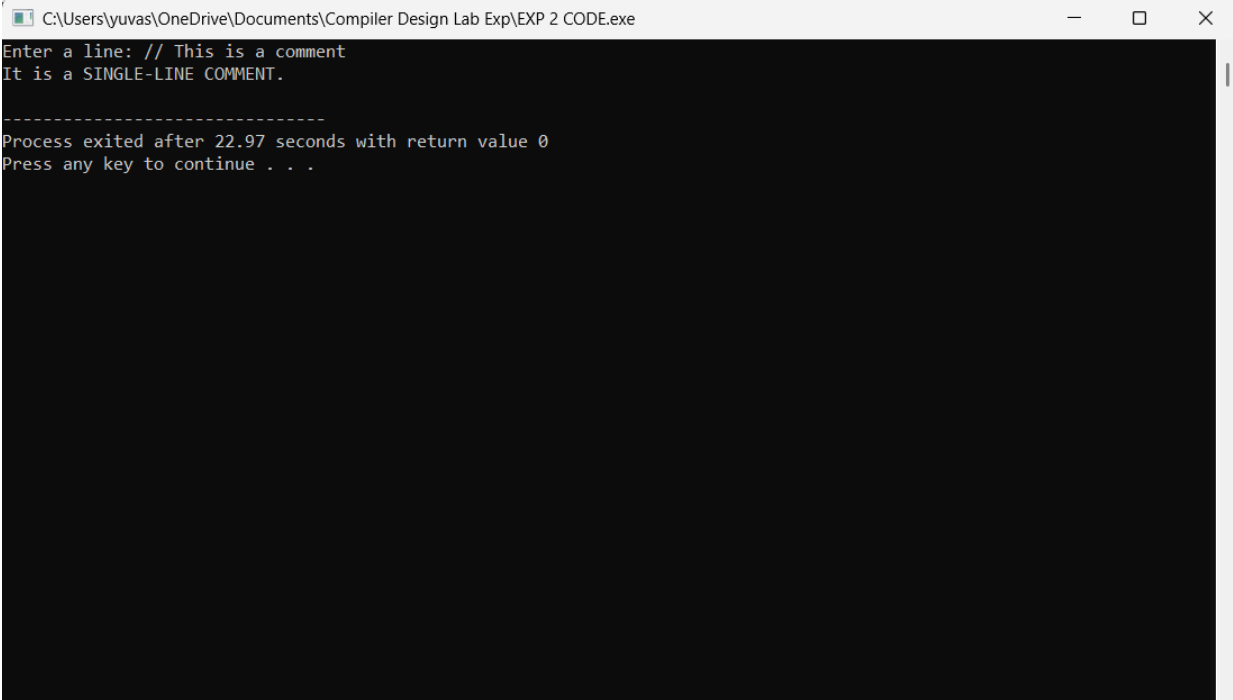
        continue;
    }

    /* Ignore other symbols */
    i++;
}

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    return 0;  
}
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OUTPUT :



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C:\Users\yuvas\OneDrive\Documents\Compiler Design Lab Exp\EXP 2 CODE.exe  
Enter a line: // This is a comment  
It is a SINGLE-LINE COMMENT.  
-----  
Process exited after 22.97 seconds with return value 0  
Press any key to continue . . .
```